

Peter A. Lilley

Software Engineer - Local LLM Inference, Language-to-Geometry Systems, Production Tooling

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SUMMARY

Software engineer who ships production systems other professionals use daily. Built a language-to-geometry pipeline: a plain-language interface that generates 3D models and sets up engineering deliverables, with locally hosted LLM inference handling intent and deterministic C#/.NET code owning geometry construction and validation. Also builds OCR ingestion and automation that cut repetitive drafting from 8-12 hours to about 30 seconds per cycle. Works in C#, Python, C++17, TypeScript, and Dart.

SKILLS

Languages: C#/.NET, Python, C++17, Dart/Flutter, TypeScript/JavaScript, SQL, LISP, VBA

AI and local inference: Ollama, LM Studio, Open WebUI, retrieval over curated corpora, Google Gemini API pipelines, Tesseract OCR, whisper.cpp speech-to-text, prompt engineering, Levenshtein matching

Engineering and data: Linear algebra and matrix transforms, 3D projection and depth sorting, graph and dependency processing, SQLite, Firebase, Supabase, Postgres, Express, React, Three.js, Git, CMake

Domain systems and quality: Civil 3D and AutoCAD API plugins, Revit/Dynamo automation, structured entity metadata, ARCore; geometric dimensioning and tolerancing (GD&T), coordinate measuring machines, ISO 9001, PPAP

EXPERIENCE

David Mason & Associates - Staff Technician (CAD automation / Civil 3D tools)

Jul 2024 - Present

St. Louis, MO

- Build CAD generation tooling that turns natural-language input into 3D models and drawing setup, using local LLM inference for intent and deterministic C#/.NET for geometry and validation; introduced this capability at the firm.
- Built OCR ingestion that parses existing PDF plan sets to auto-check and auto-populate them, replacing a manual mark-up loop.
- Automate repetitive drafting with Python and Dynamo, cutting cycles from 8-12 hours to about 30 seconds and reducing errors reaching licensed-engineer review by roughly 25%.
- Design typed metadata schemas carried on model entities, and coordinate standards automation across civil, mechanical, plumbing, and architectural disciplines.

Component Bar Products - Quality Engineer - Manufacturing / Precision Machining

Apr 2023 - Jun 2024

O'Fallon, MO

- Programmed and operated coordinate measuring machines, verifying precision-machined parts to ± 0.001 inch or tighter against GD&T specifications.
- Validated dimensional tolerances against CAD models and customer prints so machining problems surfaced as geometry mismatches, not downstream scrap.
- Enforced ISO 9001 across production lines, integrated Production Part Approval Process documentation, and managed calibration and traceability of metrology assets.

Menard, Inc. - Assistant Department Manager, Building Materials - and Pro Sales

Jun 2020 - Mar 2023

Manchester-Ballwin, MO

- Produced quantity takeoffs and commercial and residential material estimates under deadline, and handled contractor-facing pro sales that turned vague requirements into exact material lists.
- Supervised department staff in a high-volume Building Materials department, and ran inventory, pricing data, and freight quality control.

Heideman & Associates, Inc. - Revit/CAD Technician - Mechanical/Plumbing

May 2018 - Apr 2019

Fenton, MO

- Produced mechanical and plumbing construction documents and as-built models in Revit and AutoCAD, auditing standards for cross-discipline consistency.
- Built automated data libraries for building-system objects and standardized hundreds of outdated details.

PROJECTS

Digital Twin Pro

Personal product, Free Parameter LLC | Flutter/Dart 3, SQLite, Firebase, Google Gemini API, ARCore | github.com/freeParameterized/digital-twin-pro

- Designed, funded, and shipped a cross-platform 3D inventory application to a Google Play beta, plus a Windows desktop build.
- Wrote the 3D renderer by hand instead of using a game engine: painter's-algorithm scene rendering with a Z-sorted draw queue and custom projection.
- Added photo-based item detection and voice entry via a paid Gemini API, SQLite persistence with a move audit log, QR booklets, and CSV/JSON export.

CAD integration bridge

C++17, Dear ImGui, Windows COM | local repository

- Built a C++17 Dear ImGui host that COM-bridges BricsCAD, AutoCAD, and Civil 3D, discovers installed SDKs at runtime, and hot-loads LISP, C#, and BRX plugins (~14k lines).

Living resume (this site)

React, TypeScript, Three.js, Express, Ollama

- Built a React Three Fiber portfolio over an Express API that retrieves from a curated corpus and answers through a local model, with on-device speech, degrading to extractive answers when no model runs.

EDUCATION

Coursework toward Latin American Studies with Technical Applications - Missouri University of Science and Technology

2019 - 2020

Rolla, MO

Object-oriented C++ and linear algebra coursework. Technical fluency in Spanish.

A.A.S., Building Systems Engineering Technology - Ranken Technical College

2019

Wentzville, MO